



ArrestorIQ

Cross-Platform Flame Arrestor Data Management & Performance Evaluation Platform



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Abstract

Flame arrestors are safety-critical devices used in industrial pipelines to prevent flame propagation. Evaluating their performance requires comparing **pressure drop data** across product configurations, test conditions, and **computational fluid dynamics (CFD)** simulations, which is a process that was previously fragmented across disconnected spreadsheets with no standardized schema or validation.

ArrestorIQ is a **cross-platform desktop and mobile** application built for Emerson that unifies arrestor product data management under a single validated platform. The application ingests and validates test, CFD, and bill-of-materials data via the **Emerson Data Interchange (EDI) standard**, implemented as a purpose-built **Haxe-compiled parsing engine** that runs natively in both JavaScript and Python. Engineers can import new data through a drag-and-drop interface, browse a catalog of product configurations, compare **regression models** across multiple products, and inspect the full hierarchical **BOM tree** with deeplinks into Emerson's **IAS system**. A fully automated **CI/CD pipeline** integrated with Emerson's **Azure DevOps** infrastructure triggers verified releases from git tags, making the tool self-sustaining post-handoff.

Keywords: Cross-Platform Engineering, Regression Analysis, CFD Simulation, Haxe, Electron, React Native, Azure DevOps

Tech Stack



Architecture

EDI Ingestion Engine: Parses test and CFD simulation CSV files, normalizing all measurements into a consistent format.

Offline Product Catalog: Embeds the full product catalog and test data directly in the app.

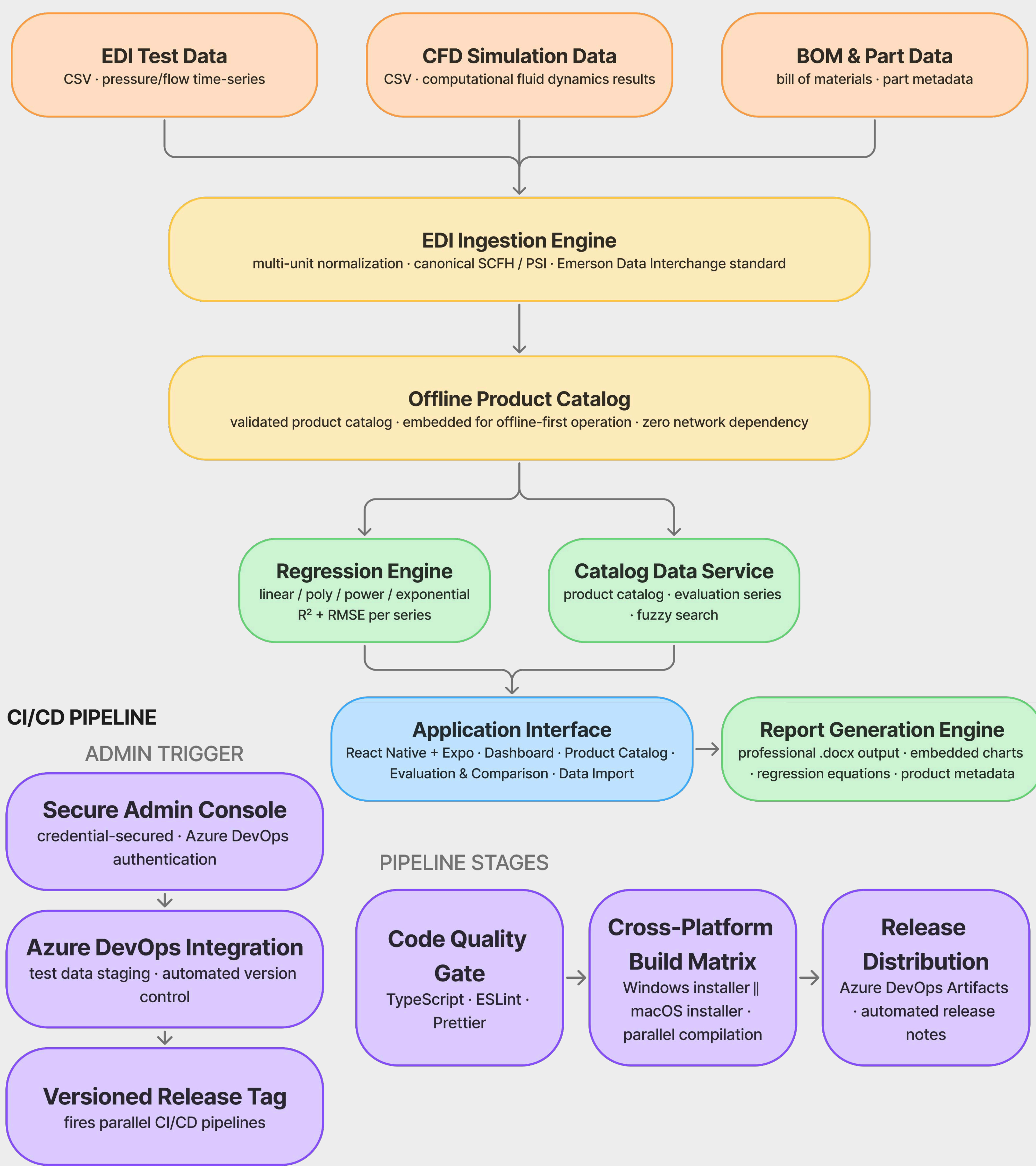
Regression Engine: Fits four model types to pressure drop data and scores each fit for cross-product comparison.

Catalog Data Service: Provides all screens with fast, searchable access to the full product catalog.

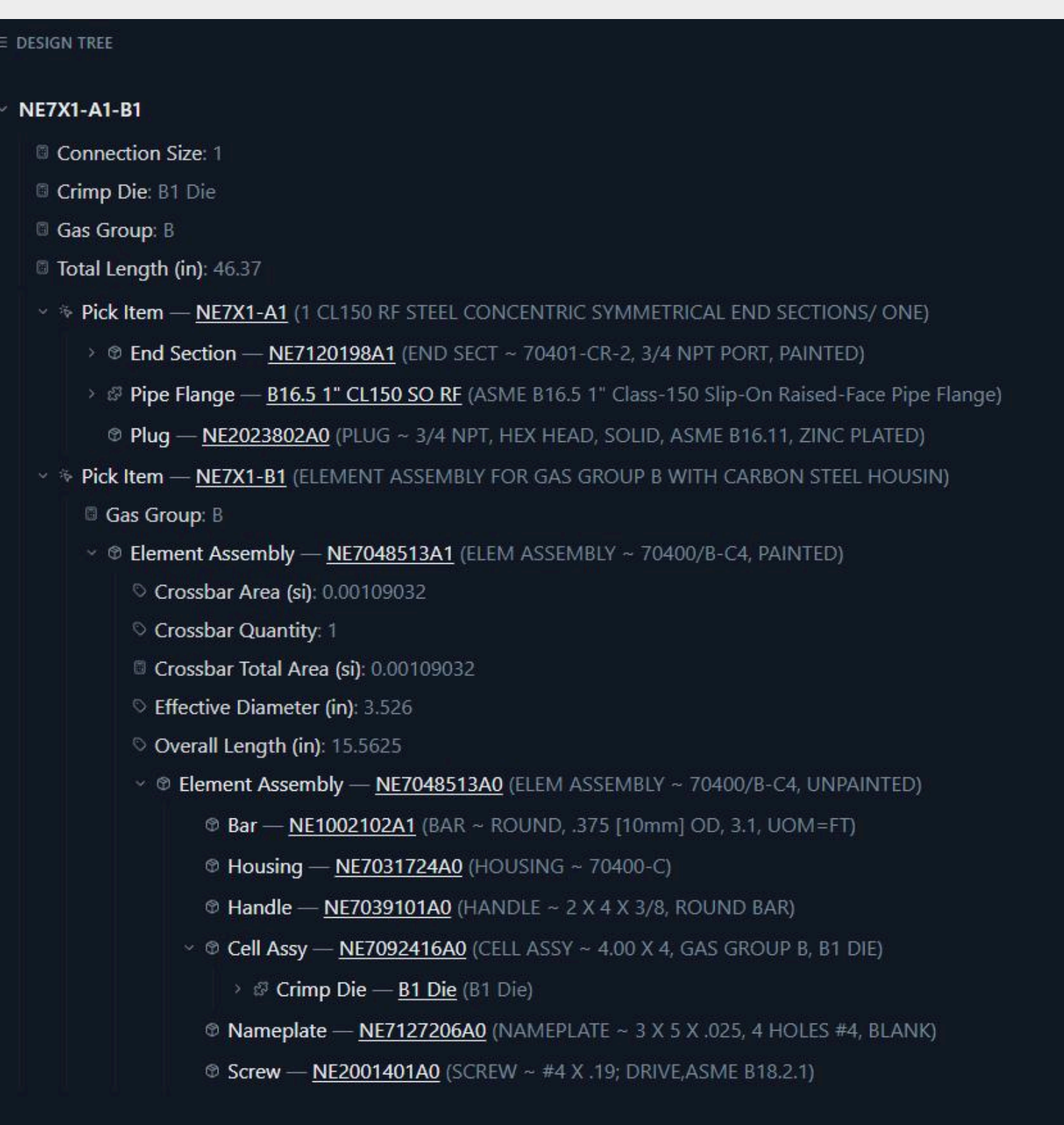
Application Interface: Dashboard, product catalog, multi-product regression comparison with BOM tree and Emerson IAS links, and drag-and-drop data import.

Report Generation Engine: Exports analysis results as a formatted Word document with charts and product specs.

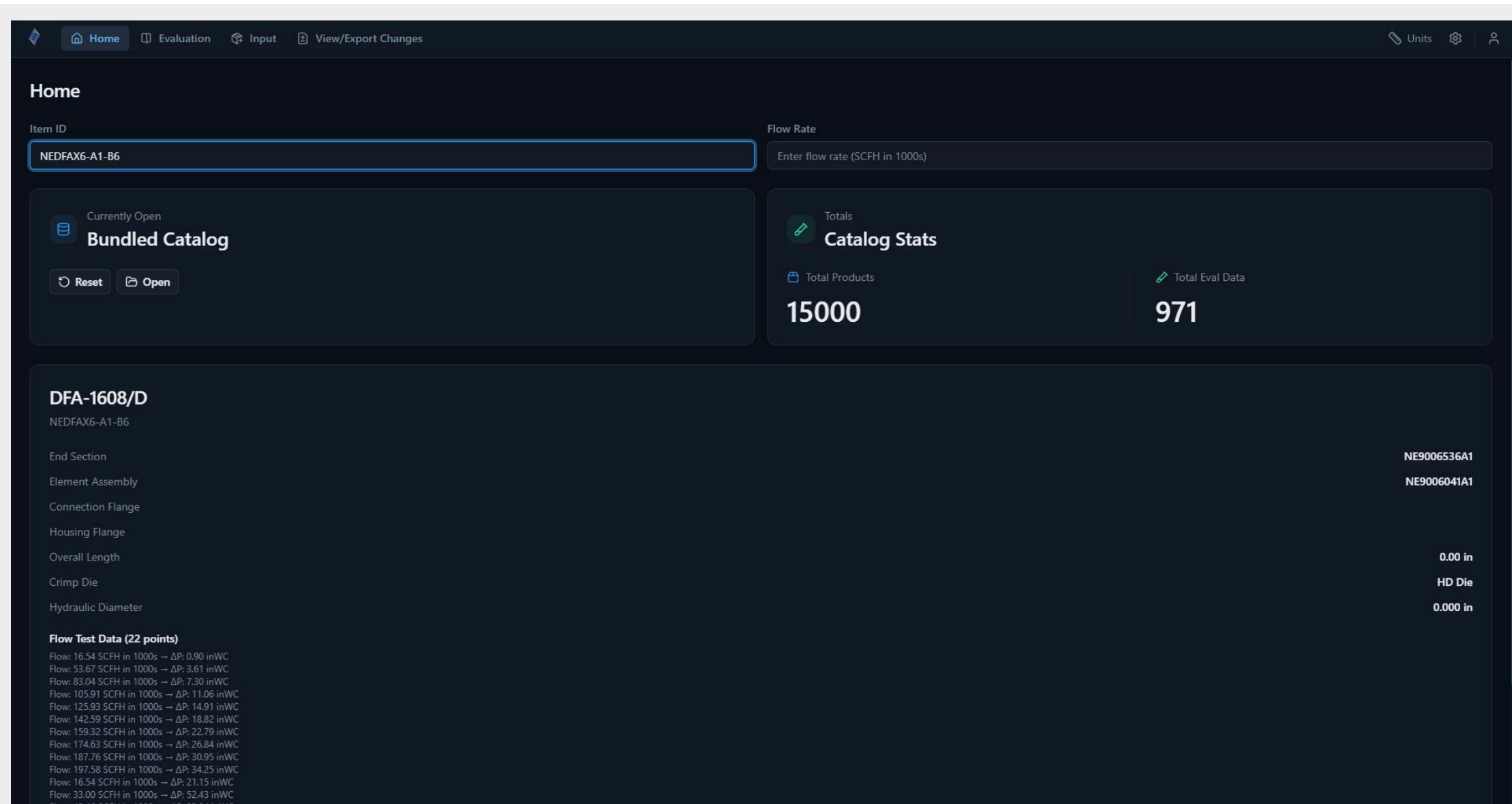
Automated Release Pipeline: Engineers import test and BOM data directly in-app. Version tags trigger automated builds and publish verified installers.



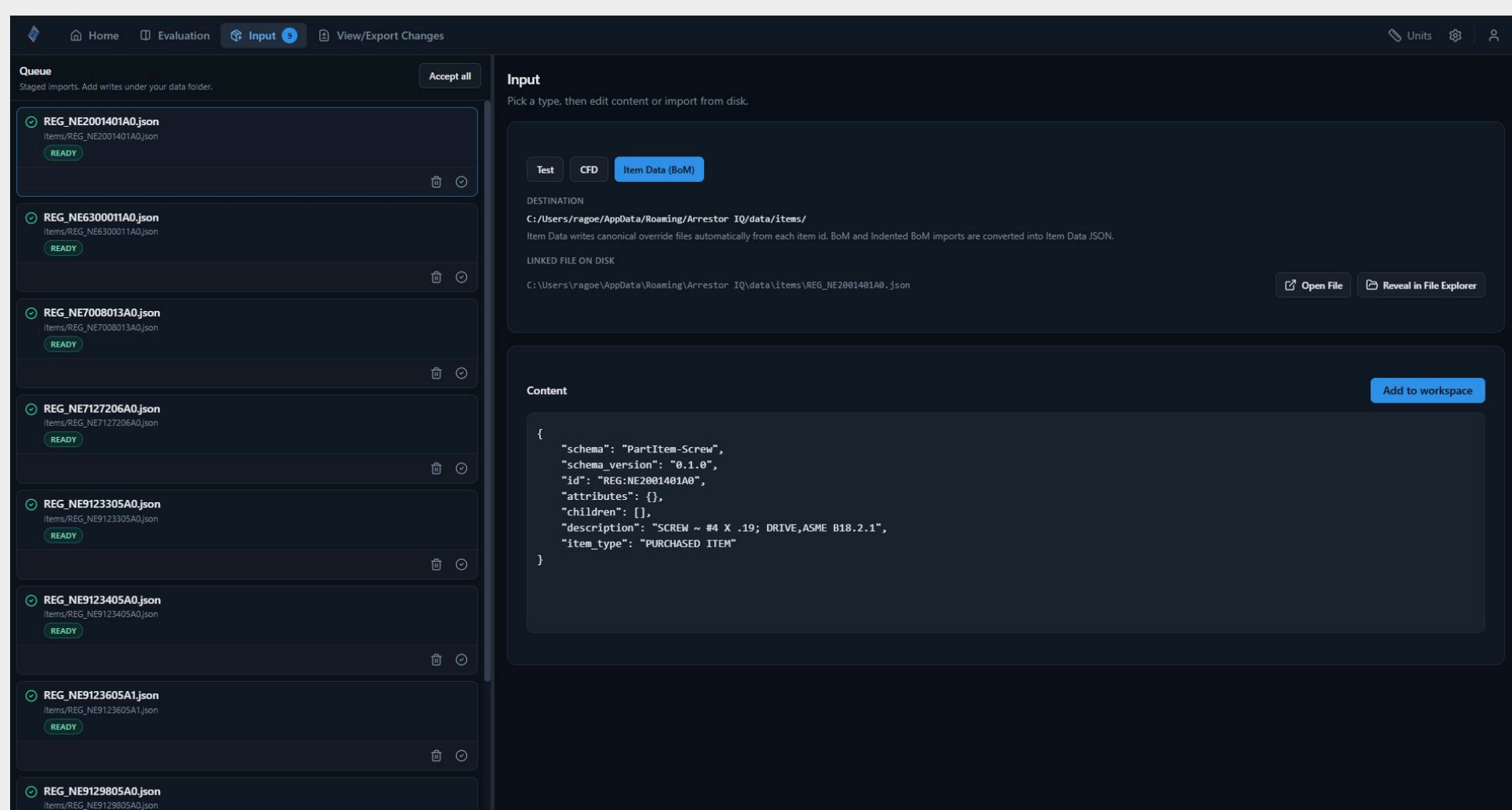
Features



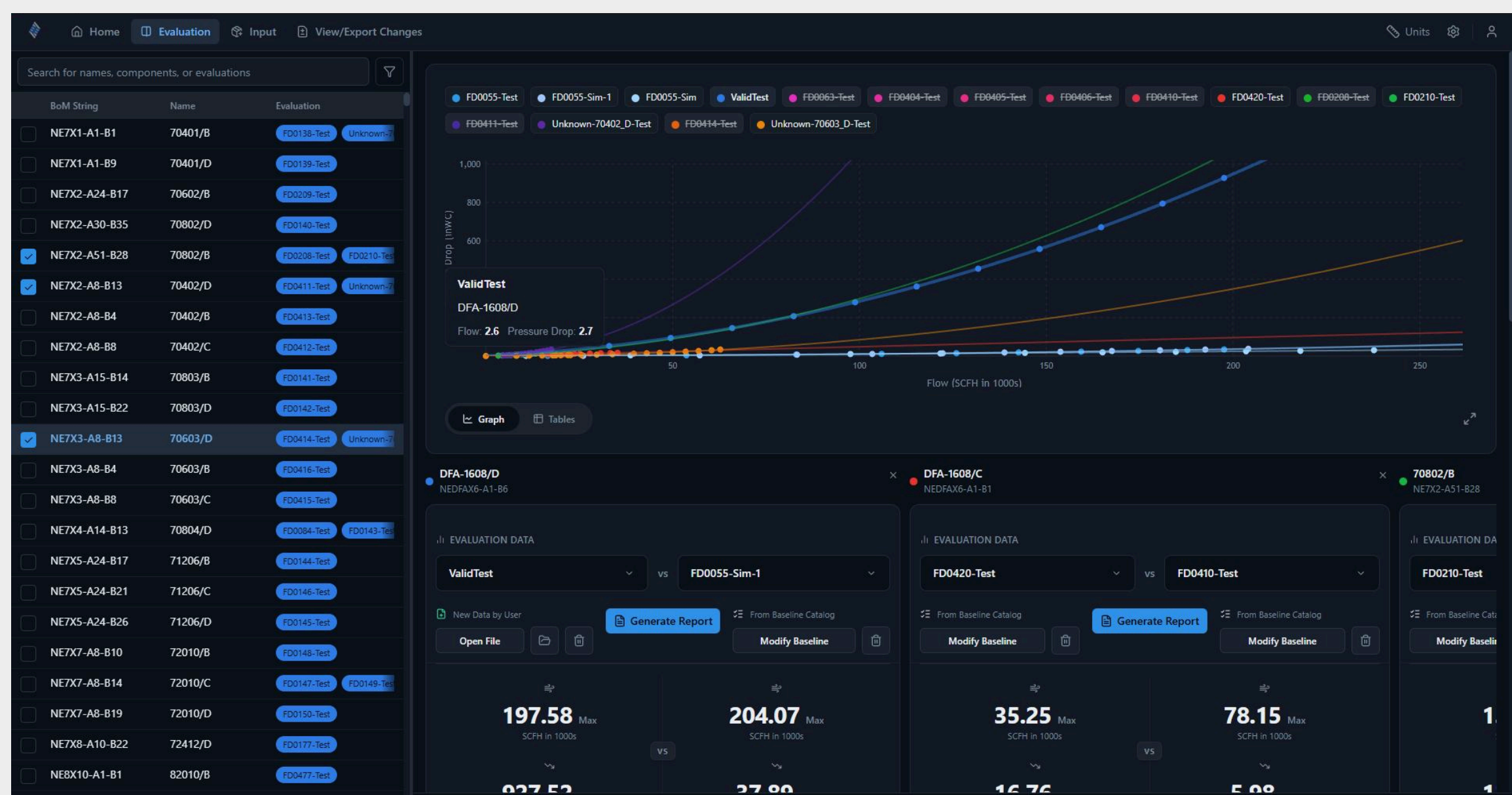
BOM Tree
Hierarchical part breakdown with Emerson IAS links



Home/Dashboard
Product lookup and test coverage at a glance



Data Import
Drag-and-drop ingestion with live validation



Evaluation & Catalog

Filter and select products, compare test vs. simulation curves, view regression fits with R^2 and RMSE scoring, and inspect the full component BOM tree

Key Statistics



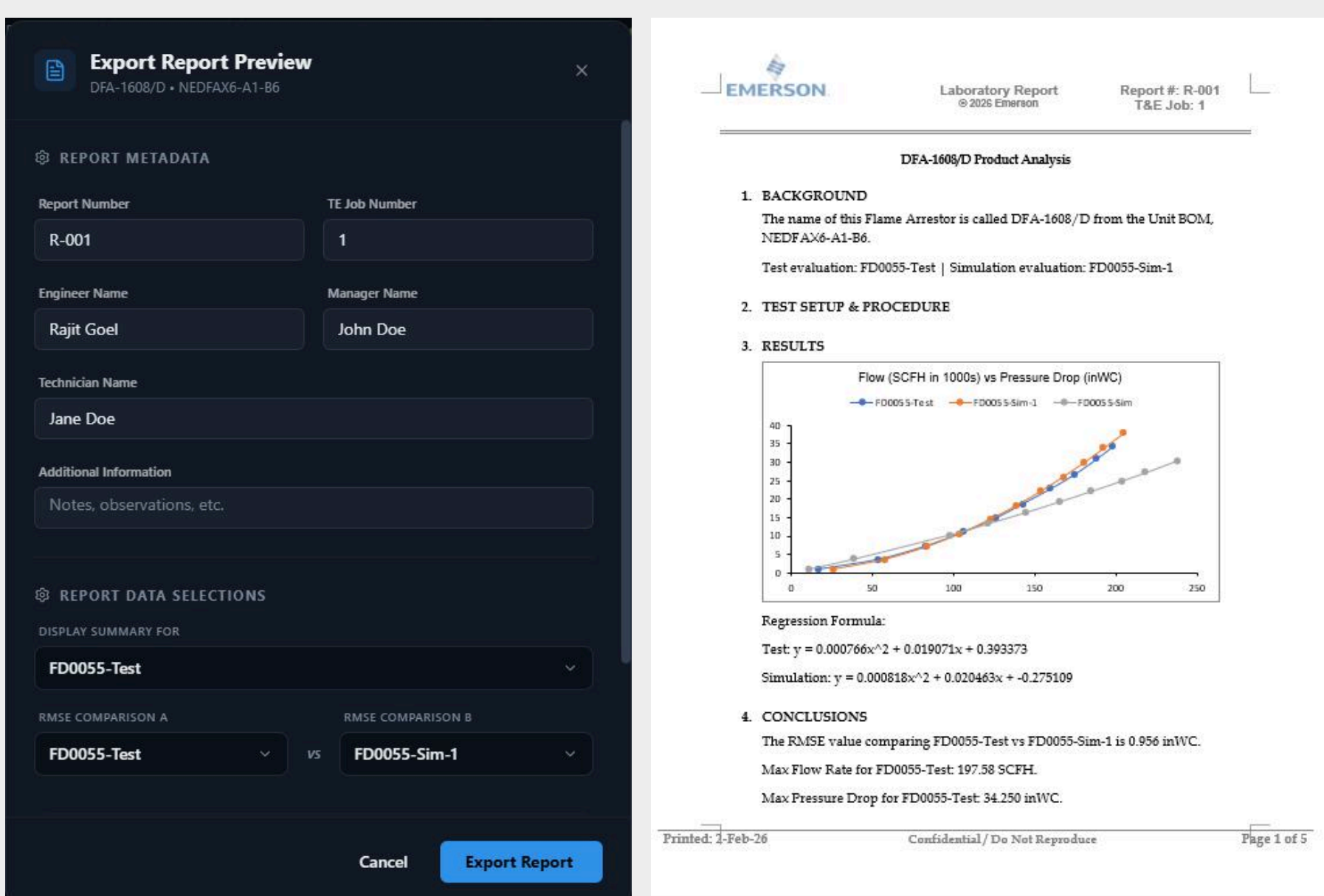
Impact and Results

- Replaces disconnected spreadsheets with a single validated platform for the entire team
- Engineers compare multiple arrestor configurations in seconds instead of hours
- Drag-and-drop data import eliminates manual data preparation steps
- Direct links to Emerson's part database from within the app
- Runs on Windows, macOS, Linux, iOS, and Android from one shared codebase
- Self-sustaining after handoff so Emerson ships updates without developer involvement
- Structured data validation catches formatting errors before they enter the system
- Exportable Word reports replace manual chart-and-paste documentation workflows

Conclusion

ArrestorIQ replaces fragmented spreadsheet workflows with a purpose-built platform for Emerson's flame arrestor product line. By unifying data ingestion, multi-product regression analysis, and direct access to engineering master data, it gives engineers faster insights with greater confidence and stays maintainable long after the project concludes.

Looking ahead, future iterations could introduce predictive performance modeling using machine learning trained on historical test data, enabling pressure drop predictions for untested configurations and speeding up the design process. Custom report templates would allow individual departments to export documents that automatically match their own documentation standards. Expanded audit trail support including batch imports and history tracking across products and evaluations would make the platform ready for larger engineering teams.



Report Export

One-click export to a formatted Word report